Validation and the likelihood surface

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Overview

The paper fits the model to the survey of 2008–2019 and holds back 2020–2023 to test it. This vignette reproduces that test at the paper's best parameter set (Table 4), maps the forecast, and draws the likelihood on a regular grid, PEOC's actions 5 and 8.

```
using AshDieback
using Plots
```

The best parameter set

The paper's θmax (eq. 14) is the sample of its AMIS run with the highest likelihood:

```
data = load_peoc_data()
θmax = AshDiebackParameters(; r_symp = 1000.0, r_allee = 0.0, beta_0 = 23.26, beta_1 = 0.0,
                            D = 18.51, S_sat = 92.0, gamma = 21.50001, C_pers = 1.0,
                            C_init = 0.00855, kappa = 0.056)
prob = AshDiebackProblem(data, θmax)
loglikelihood(prob)
```

```
-24717.21496294487
```

Forecasting beyond the fit

Changing `final_year` extends the simulation with the observed climate. Nothing is refitted:

```
forecast = remake(prob; settings = ModelSettings(; final_year = 2023))
sol = solve(forecast)
plot(plot(sol; year = 2019), plot(sol; year = 2023); layout = (1, 2), size = (900, 380),
     clims = (0, 1))
```

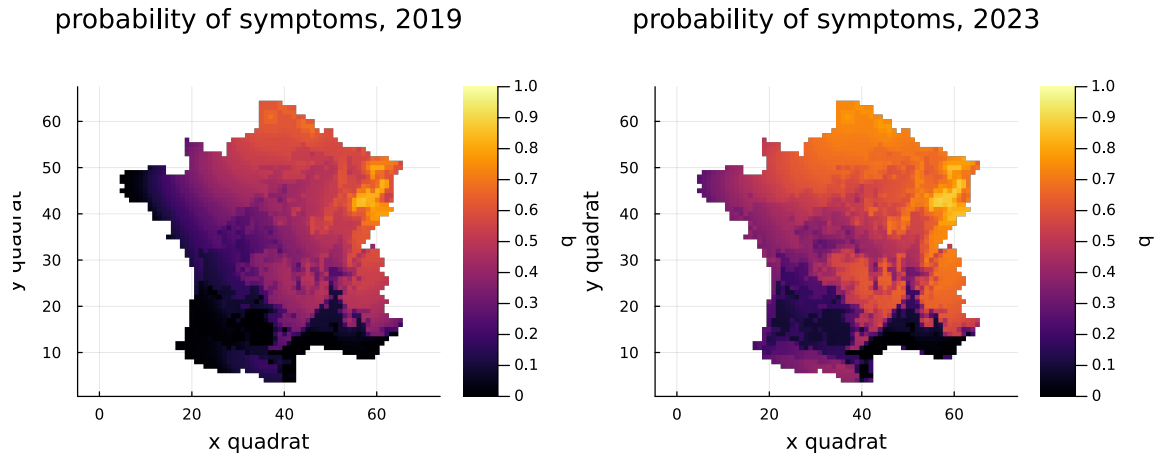


Table 4

`rmse` compares each plot visit with the probability of symptoms in its quadrat, each year and cumulatively:

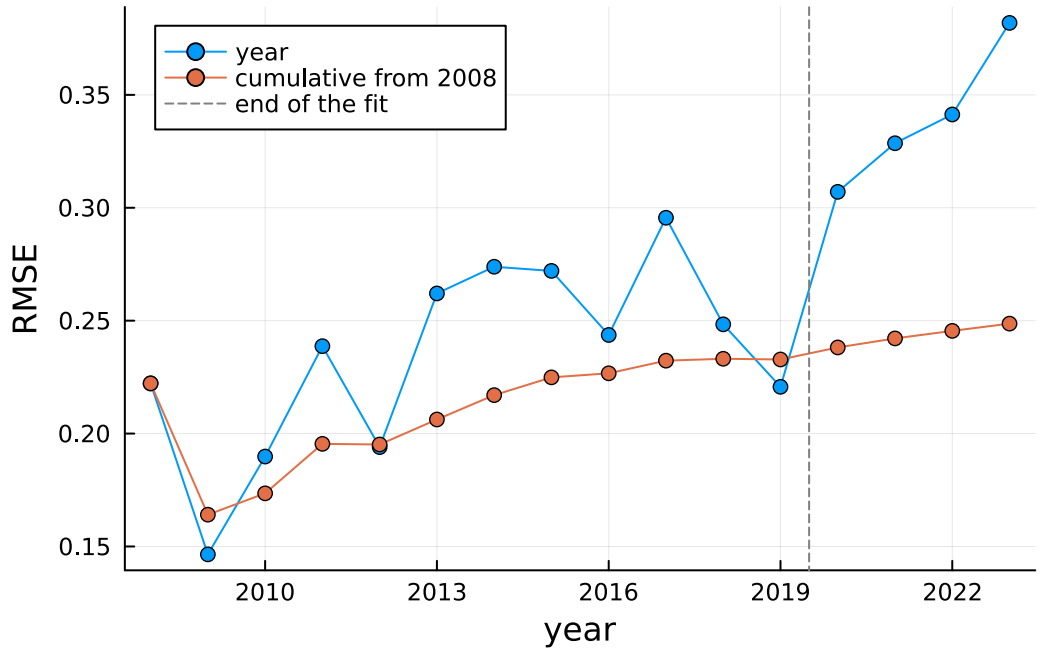
```
t = rmse(sol)
```

```
RMSETable: year, observations, RMSE of the year, RMSE from 2008
  2008   183  0.2222  0.2222
  2009   753  0.1465  0.1641
  2010   504  0.1898  0.1735
  2011   620  0.2387  0.1954
  2012   515  0.1940  0.1952
  2013   438  0.2621  0.2062
  2014   493  0.2739  0.2170
  2015   520  0.2720  0.2249
  2016   410  0.2437  0.2267
  2017   341  0.2956  0.2323
  2018   254  0.2484  0.2331
  2019   132  0.2207  0.2328
```

```
  2020   345  0.3070  0.2381
  2021   214  0.3286  0.2421
  2022   166  0.3413  0.2455
  2023   111  0.3819  0.2487
```

All of these are within one unit in the fourth decimal of the paper's table. Two differ in that digit: the cumulative value for 2012 (0.1952 against 0.1951) and the RMSE of 2020 (0.3070 against 0.3071). The printed θmax is rounded, so such differences are to be expected. The paper's table also has one more plot in 2020 and in 2021 than the released data. The years from 2020 were not used in the fit, and the error rises there, as the paper reports.

```
plot(t.years, [t.annual t.cumulative]; label = ["year" "cumulative from 2008"],
     marker = :circle, xlabel = "year", ylabel = "RMSE")
vline!([2019.5]; label = "end of the fit", linestyle = :dash, color = :gray)
```



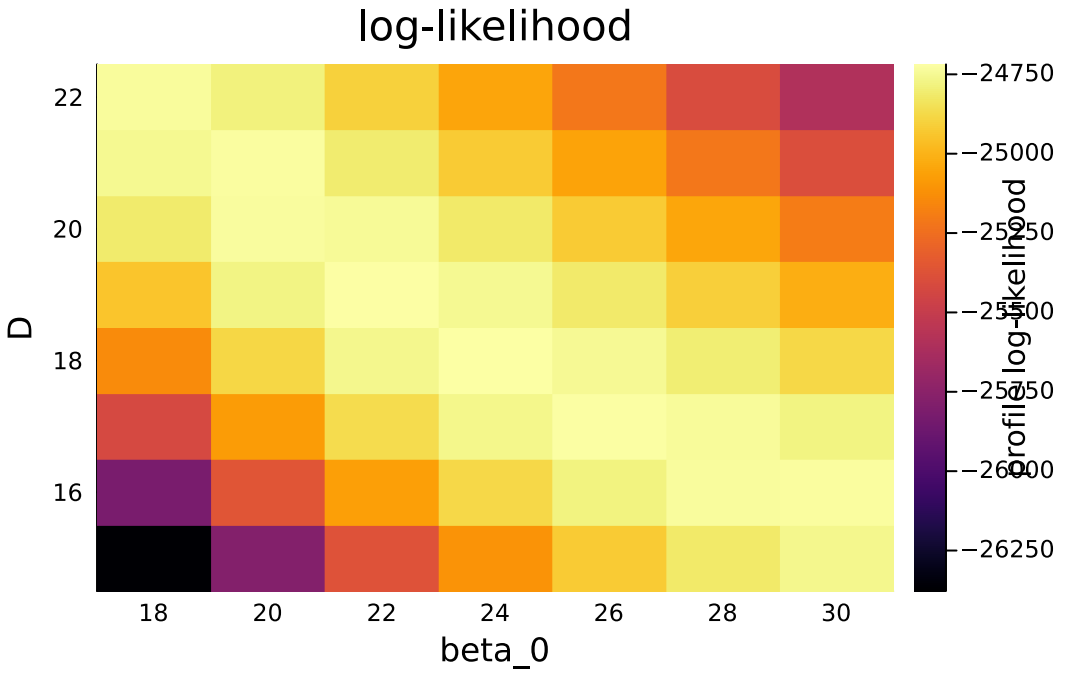
The likelihood on a grid

`likelihood_grid` evaluates the log-likelihood on a Cartesian product of parameter values, holding the rest at their values in the problem. It uses threads when Julia has them. Here are β_0 and D, the strongly correlated pair:

```
grid = likelihood_grid(prob, (; beta_0 = range(18, 30; length = 7),
                                D = range(15, 22; length = 8)))
best_parameters(grid)
```

```
([1000.0, 0.0, 24.0, 0.0, 18.0, 92.0, 21.50001, 1.0, 0.00855, 0.056], -
24718.651008031204)
```

```
plot(grid; x = :beta_0, y = :D, title = "log-likelihood")
```



The ridge from high β_0 and low D to low β_0 and high D is the correlation of -0.97 in the paper's Table 3: more spores are offset by slower diffusion.

With more than two axes, `plot` shows the profile, maximised over the others, and `profile_likelihood` returns it. PEOC's own grid (`peoc_grid_axes()`) has 630 000 points, which is several hours' work on one machine.